MA1522 Linear Algebra for Computing

Lesson 11: Coordinates and Transition Matrices

Intuition:

If $V \subseteq \mathbb{R}^n$ is a subspace of dimension k.

then V "looks the same as" $\mathbb{R}^k$.

If we fix an ordered basis $\{u_1, \ldots, u_k\} = S$ of V.

then we obtain $V \longrightarrow \mathbb{R}^k$ "isomorphism".

$v \longmapsto [v]_S$.

Submission

- You only need to submit your attempts at those labelled **Question 1** to **Question 3**.
- If you have written your answer on A4 paper, scan or take pictures of your work (make sure the images can be read clearly). Merge all your images into one PDF file, in page order.
- If you have written or typed your answer digitally, print it into a PDF file.
- Name the PDF file by `StudentNo_lesson11` (e.g. `A1234567Z_lesson11.pdf`).
- Submit the PDF file to Canvas assignments, **Participation(L1): Lesson 11**.
- Deadline: today, **12:30 PM**.
- If you do not know how to upload your submission, come to ask me after class.

Pollev



Section 3.6 Coordinates Relative to a Basis

Fix $S = \{\mathbf{u}_1, \ldots, \mathbf{u}_k\}$. $V \longrightarrow \mathbb{R}^k$.

$\mathbf{v} \longmapsto [\mathbf{v}]_S$.

Definition

Let $S = \{\mathbf{u}_1, \ldots, \mathbf{u}_k\}$ be a basis for V, a subspace of $\mathbb{R}^n$, and let $\mathbf{v} \in V$ have the unique expression $\mathbf{v} = c_1\mathbf{u}_1 + \cdots + c_k\mathbf{u}_k$. The vector $[\mathbf{v}]_S = (c_1, \ldots, c_k)^T \in \mathbb{R}^k$ is called the **coordinates of v relative to the basis** S.

Algorithm

To find $[\mathbf{v}]_S$, solve the linear system $\left(\mathbf{u}_1\ \mathbf{u}_2\ \cdots\ \mathbf{u}_k \mid \mathbf{v}\right)$.

Note that $\mathbf{v} \in \mathbb{R}^n$ but $[\mathbf{v}]_S \in \mathbb{R}^k$, and that $[\mathbf{v}]_S$ depends on the ordering of S.

Example 1

Let $S = \{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$ be a basis for a subspace $V \subseteq \mathbb{R}^5$, and let $\mathbf{v} \in V$ be such that

$$\operatorname{rref}\left(\begin{array}{ccc|c} \mathbf{u}_1 & \mathbf{u}_2 & \mathbf{u}_3 & \mathbf{v} \end{array}\right) = \left(\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & -5 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right).$$

$c_1 = 1$.

$c_2 = -5$.

$c_3 = 0$.

What is $[\mathbf{v}]_S$?

$c_1 u_1 + c_2 u_2 + c_3 u_3 = v$.

$$\begin{pmatrix} u_1 & u_2 & u_3 \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix} = v \quad .$$

$v = 1 \cdot u_1 + (-5) \cdot u_2 + 0 \, u_3$.

$$\Rightarrow [v]_S = \begin{pmatrix} 1 \\ -5 \\ 0 \end{pmatrix} .$$

Question 1

Let V be the subspace of $\mathbb{R}^4$ with basis $S = \{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$, where

$$\mathbf{u}_1 = \begin{pmatrix} 1 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \qquad \mathbf{u}_2 = \begin{pmatrix} 0 \\ 1 \\ 1 \\ 0 \end{pmatrix}, \qquad \mathbf{u}_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 1 \end{pmatrix}.$$

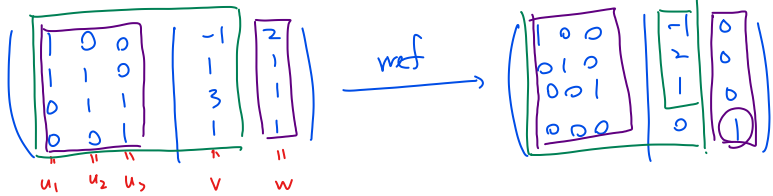
(a) Show that $\mathbf{v} = (-1, 1, 3, 1)^T$ lies in V, and find $[\mathbf{v}]_S$.

(b) Show that $\mathbf{w} = (2, 1, 1, 1)^T$ does not lie in V.

(c) Suppose $\mathbf{x} \in V$ has $[\mathbf{x}]_S = (2, -1, 3)^T$. Find $\mathbf{x}$.

$$\begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix} = (2 \ -1 \ 3)^T$$

Question 1 — solution



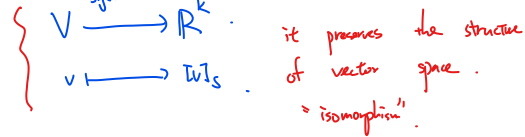
a) $[v]_S = \begin{pmatrix} -1 \\ 2 \\ 1 \end{pmatrix}$

b) The linear system $c_1u_1 + c_2u_2 + c_3u_3 = w$ is inconsistent. $\Rightarrow$ $w \notin V = \text{span}(S)$.

c) Suppose $[x]_S = \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix}$, What is x?

$$x = 2u_1 + (-1)u_2 + 3u_3 = 2\begin{pmatrix} 1 \\ 1 \\ 0 \\ 0 \end{pmatrix} + (-1)\begin{pmatrix} 0 \\ 1 \\ 1 \\ 0 \end{pmatrix} + 3\begin{pmatrix} 0 \\ 0 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \\ 2 \\ 3 \end{pmatrix}$$

Section 3.6 Properties of Coordinates



Proposition

Let S be a basis for a subspace V of $\mathbb{R}^n$. Then coordinates **preserve linear combinations**: for all $\mathbf{v}_1, \ldots, \mathbf{v}_m \in V$ and all scalars $c_1, \ldots, c_m$,

$$[\, c_1\mathbf{v}_1 + \cdots + c_m\mathbf{v}_m \,]_S = c_1[\mathbf{v}_1]_S + \cdots + c_m[\mathbf{v}_m]_S.$$

Moreover the correspondence $\mathbf{v} \leftrightarrow [\mathbf{v}]_S$ is one-to-one: if $[\mathbf{x}]_S = [\mathbf{y}]_S$ then $\mathbf{x} = \mathbf{y}$, since the expression of a vector in a basis is unique.

Example 2

True or false? Let $S = \{\mathbf{u}_1, \ldots, \mathbf{u}_k\}$ be a basis for $V \subseteq \mathbb{R}^n$, and let $\mathbf{w}_1, \ldots, \mathbf{w}_l$ and $\mathbf{v}$ be vectors in V.

1. $\{\mathbf{w}_1, \ldots, \mathbf{w}_l\}$ is linearly independent if and only if $\{[\mathbf{w}_1]_S, \ldots, [\mathbf{w}_l]_S\}$ is linearly independent.

2. $\mathbf{v} \in \text{span}\{\mathbf{w}_1, \ldots, \mathbf{w}_l\}$ if and only if $[\mathbf{v}]_S \in \text{span}\{[\mathbf{w}_1]_S, \ldots, [\mathbf{w}_l]_S\}$.

($\Leftarrow$).

1. Suppose $c_1 w_1 + \cdots + c_l w_l = 0$.

$$\begin{pmatrix} 0 \\ \vdots \\ 0 \end{pmatrix} = [0]_S = [c_1 w_1 + \cdots + c_l w_l]_S \overset{\text{Prop}}{=\!=} c_1 [w_1]_S + \cdots + c_l [w_l]_S$$

$\{[w_1]_S, \ldots, [w_l]_S\}$ is l.I.

$c_1 = c_2 = \cdots = c_l = 0 \implies \{w_1, \ldots, w_l\}$ is l.I.

2. $v = c_1 w_1 + \cdots + c_l w_l \iff [v]_S = c_1 [w_1]_S + \cdots + c_l [w_l]_S$.

Question 2

Let $S = \{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$ be the basis for $V \subseteq \mathbb{R}^4$ ~~of Question 1,~~ and let $T = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\} \subseteq V$ be such that

$$\begin{pmatrix} [\mathbf{v}_1]_S & [\mathbf{v}_2]_S & [\mathbf{v}_3]_S \end{pmatrix} = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 1 & 2 \end{pmatrix}.$$

(a) Show that T is linearly independent. $\Leftrightarrow \{[\mathbf{v}_1]_S, [\mathbf{v}_2]_S, [\mathbf{v}_3]_S\}$ is L.I.

(b) Express $\mathbf{u}_1$ as a linear combination of the vectors in T.

Question 2 — solution

a) $\left([v_1]_S \;\; [v_2]_S \;\; [v_3]_S \right) = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 1 & 2 \end{pmatrix} = A$

3 vectors in $\mathbb{R}^3$.

So $\{[v_1]_S, [v_2]_S, [v_3]_S\}$ is l.I. $\iff$ A is invertible $\iff$ $\det(A) \neq 0$.

This holds since $\det(A) = 1$.

$\Updownarrow$ Example 2.

$\{v_1, v_2, v_3\}$ is l.I.

$\left(\iff \operatorname{span}\{[v_1]_S, [v_2]_S, [v_3]_S\} = \mathbb{R}^3. \right.$

$\left. \iff \operatorname{span}(\{v_1, v_2, v_3\}) = V. \right)$ Example 2.

b) Express u_1 as a linear combination of T.

$u_1 = c_1 v_1 + c_2 v_2 + c_3 v_3$

$\Updownarrow$ Proposition

$\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = [u_1]_S = c_1 [v_1]_S + c_2 [v_2]_S + c_3 [v_3]_S$

these are given.

$u_1 = 1 \cdot u_1 + 0 \cdot u_2 + 0 \cdot u_3$

$\left(\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 0 \\ 1 & 1 & 2 & 0 \end{array} \right)$ ref $\left(\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & -1 \end{array} \right)$

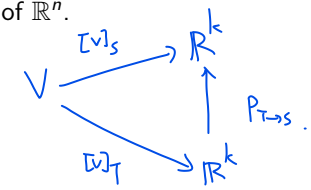
$\implies \begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix}.$

$\implies u_1 = 1 \cdot v_1 + 1 \cdot v_2 + (-1) \cdot v_3.$

Section 3.8 Transition Matrices

Let $S = \{\mathbf{u}_1, \ldots, \mathbf{u}_k\}$ and $T = \{\mathbf{v}_1, \ldots, \mathbf{v}_k\}$ be bases for a subspace V of $\mathbb{R}^n$.

$[\mathbf{v}]_S$ $[\mathbf{v}]_T$.



Definition

The **transition matrix from T to S** is $P_{T\to S} = \big([\mathbf{v}_1]_S \; [\mathbf{v}_2]_S \; \cdots \; [\mathbf{v}_k]_S\big)$.

Theorem

For every $\mathbf{w} \in V$, $[\mathbf{w}]_S = P[\mathbf{w}]_T$.

Algorithm

$\big(\mathbf{u}_1 \cdots \mathbf{u}_k \mid \mathbf{v}_1 \cdots \mathbf{v}_k\big) \longrightarrow \begin{pmatrix} I_k & P \\ 0 & 0 \end{pmatrix}$: the basis you want to land in goes on the left.

Question 3

As in Question 2, let $S = \{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$ be a basis for $V \subseteq \mathbb{R}^4$ and let $T = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\} \subseteq V$ be such that

$$\begin{pmatrix} [\mathbf{v}_1]_S & [\mathbf{v}_2]_S & [\mathbf{v}_3]_S \end{pmatrix} = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 1 & 2 \end{pmatrix}.$$

$P_{T \to S} = \begin{pmatrix} [v_1]_S & [v_2]_S & [v_3]_S \end{pmatrix}$ definition.

(a) Write down the transition matrix P from T to S.

(b) Find the transition matrix Q from S to T.

(c) Compute PQ, and say what it tells you about P and Q.

Question 3 — solution: (a) and (b)

b) Find the transition matrix $Q_{S\to T} = \left([u_1]_T \; [u_2]_T \; [u_3]_T \right)$

we have computed this in Question 2 (b).

$P_{T\to S}$

$$\left(\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 1 & 2 & 3 & 0 & 1 & 0 \\ 1 & 1 & 2 & 0 & 0 & 1 \end{array} \right)$$

rref

$$\left(\begin{array}{ccc|ccc} 1 & & & 1 & -1 & 1 \\ & 1 & & 1 & 1 & -2 \\ & & 1 & -1 & 0 & 1 \end{array} \right)$$

$Q_{S\to T}$

$[u_1]_S \quad [u_2]_S \quad [u_3]_S$

$$\begin{cases} u_1 = \underline{1}\cdot u_1 + \underline{0}\cdot u_2 + \underline{0}\cdot u_3 \\ u_2 = \underline{0}\cdot u_1 + \underline{1}\cdot u_2 + \underline{0}\cdot u_3 \\ u_3 = \underline{0}\cdot u_1 + \underline{0}\cdot u_2 + \underline{1}\cdot u_3 \end{cases}$$

$$\begin{cases} u_1 = 1\cdot v_1 + 1\cdot v_2 + (-1)\cdot v_3 \\ u_2 = -1 \quad\quad 1 \quad\quad 0 \\ u_3 = 1 \quad\quad -2 \quad\quad 1 \end{cases}$$

$$\Longrightarrow \left([u_1]_T \; [u_2]_T \; [u_3]_T \right) = \begin{pmatrix} 1 & -1 & 1 \\ 1 & 1 & -2 \\ -1 & 0 & 1 \end{pmatrix} = Q_{S\to T}.$$

Question 3 — solution: (c)

Compute $PQ := I_3$.

$$\left(\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & & \\ 1 & 2 & 3 & & 1 & \\ 1 & 1 & 2 & & & 1 \end{array}\right) \xrightarrow{\text{rref}} \left(\begin{array}{ccc|ccc} 1 & & & 1 & -1 & 1 \\ & 1 & & 1 & 1 & -2 \\ & & 1 & -1 & 0 & 1 \end{array}\right)$$

P $\qquad$ Q.

So by the algorithm to compute inverse, we see that $Q = P^{-1}$.

Challenge

$P_{T \to S}$.

Let S and T be bases for a subspace V of $\mathbb{R}^n$, and let P be the transition matrix from T to S.

Show that P is invertible, and that P^{-1} is the transition matrix from S to T.

$P_{S \to T}$.

You may not assume that P is invertible — that is what is being proved.

Hint: Use the theorem that

$$[w]_S = P_{T \to S} [w]_T \quad \text{for all } w \in V.$$

Next lesson

- **Sections 4.1 & 4.2 Column Space, Row Space, Nullspace, and Rank**

Please watch the video clips for these sections beforehand.